

Python Project Collection

This document showcases beginner-to-intermediate Python projects built as part of a learning journey. Most of these projects were guided by Tech With Tim's tutorials, with a few original or school-based creations.

Transaction Terminal

A simple command-line banking system that lets users:

- Check balance
- Deposit funds
- Withdraw funds

This project demonstrates the use of functions, loops, and basic user input handling.

WPM Typing Test

A terminal-based typing speed test built using the curses library.

Features:

- Randomly loads a line from a file
- Calculates real-time Words Per Minute (WPM)

This project introduces working with the curses module, file I/O, and live input handling.

Password Manager

This project includes two versions:

- Basic version: Stores account names and passwords in plain text
- Advanced version: Uses the cryptography library to encrypt and decrypt passwords

It covers file management, basic encryption, and simple user authentication.

Slot Machine Game

A command-line slot machine simulator that includes:

- Randomized spinning symbols
- Multi-line betting system
- Balance tracking and win/loss logic

This reinforces logic building, randomness, and loop-based control.

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Currency Converter

Fetches real-time currency exchange rates from currconv.com API.

Supports:

- Listing available currencies
- Checking exchange rates
- Converting currency amounts

The project teaches how to use APIs, handle JSON data, and perform real-time calculations.

Calculator UI

A functional calculator with a graphical user interface using Tkinter.

Features:

- Basic operations (+, -, ×, ÷)
- Percentage, sign change, and clear functionality
- Expression evaluation with Python's built-in logic

This combines GUI layout with mathematical logic.

Additional Projects

Graph Sketcher

A GUI-based graphing calculator that:

- Plots mathematical functions using ``eval``
- Includes zooming, clearing, and error handling
- Uses ``tkinter`` for GUI and ``math`` for evaluating expressions

Triangle Tessellation

A console-based pattern generator that:

- Draws a repeating triangular pattern based on user-defined height, rows, and columns
- Demonstrates nested loops, modular arithmetic, and string formatting

Chessboard with Turtle

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A graphical application that:

- Uses Python's `turtle` module to draw an 8x8 chessboard
- Applies nested loops and conditional color filling

Digital Digits Renderer

Creates pixelated representations of digits 0-9 using `turtle` graphics.

- Demonstrates geometric logic and procedural programming
- Digits drawn by stamp patterns for visual output

Collision Detection Module

Two versions of geometric rectangle comparison functions that:

- Check for overlapping, containment, and covering of rectangles
- Useful in 2D games or UI bounding box logic

Car Racing Game

A fun, interactive game where:

- Two cars race around a track controlled via keyboard inputs
- The track is visualized using Turtle graphics
- Demonstrates game logic, object control, event handling, and real-time visuals